LAST NAME: _____

Practice Quiz 07C — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. A student claims that every symmetric, transitive relation R on a set A is reflexive, and argues:

   > Let $a \in A$. Choose $b \in A$ with aRb. Symmetry gives bRa. Transitivity then gives aRa.

   (a) Identify the unsupported step. Verify that $R = \{(1,1)\}$ on $A = \{1,2\}$ is a counterexample by checking symmetry, transitivity, and failure of reflexivity.

   (b) Prove that the claim becomes true if one also assumes $\forall a \in A\, \exists b \in A\, (aRb)$.